

Practice set 3

Quadratics and the discriminant

Write in the spaces

- (b) A quadratic equation has solutions of $x = -\frac{2}{3}$ and $x = 4$.

Find the original equation, giving your answer in the form of $ax^2 + bx + c = 0$, where a , b , and c are whole numbers.

- (c) Consider a quadratic equation in the form $x^2 - 3kx + 2k^2 = 0$, where k is a non-zero constant.

Show that one solution is twice the other solution.

QUESTION TWO

- (a) A quadratic equation, $ax^2 + bx + c = 0$, has solutions of $\frac{1}{3}$ and $\frac{-2}{7}$.

Find the values of the integers a, b, and c.

a = _____ b = _____ c = _____

- (b) (i) What is the discriminant of the equation $2x^2 - 12x + 7 = 0$?

- (ii) Suppose $y = 2x^2 - 12x + k$, where k is a constant.

For what value of k will the equation $y = 0$ have exactly one solution?
